

(a) Decompose $\rightleftharpoons$ compose $\rightarrow$ data flow $\dashrightarrow$ score / creditu rejected

Masked loop

 $\mathcal{L} =$
 (Init, B, S)
G hidden

within a rollout

1 Rollout

y_1	a	b	c	a	b	✓
y_2	c	u	d	e	c	✗
y_3	v	d	u	e	f	✗

2 Decompose

A_1	a	b	c	a	b
A_2	c	u	d	e	c
A_3	v	d	u	e	f

across rollouts

3 Pool

a	b	u
c	d	v
e	f	

4 Filter**5 Compose**

a	b	e
d	e	f

Output
 C_{surv}

↓ composed set

↓ rollout group

(b) SFTcompose $\rightarrow$ label

Composed set

 $C_{\mathcal{L}}^{(t)}$

resample

Coverage $\geq \tau$?

yes

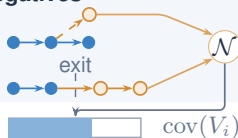
SFT labels

Fine-tune

(c) Target-free scorerBinary valid set $\mapsto \{0, 1\}$ **Candidate negatives**

off-trajectory

post-exit

**Coverage** $\text{cov}(V_i)$ **Shapley allocation**

y_1	⊗	⊗	⊗	○	○	ϕ_1
y_2	⊗	⊗	⊗	○	⊗	ϕ_2
y_3	○	○	⊗	●	⊗	ϕ_3

● unique ⊗ shared

Full reward per rollout

$$r_i = \text{cov}(V_i) + 0.3\phi_i$$

(d) RL

attribute exploration

y_1	a	b	c	a	b
y_2	c	u	d	e	c
y_3	v	d	u	e	f

pool, filtered once

a	b	c	d	e	f
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 r_1 r_2 r_3 **GRPO**
 group-normalized
 advantage
 $\rightarrow$ policy update